

SEMESTER-VI

UMT602: SIGNALS, SYSTEMS AND PROCESSING				
	L	T	P	Cr
	3	1	2*	4.0
Course Objective: The aim of this subject is to develop analytical capability of students, by which they would be able to handle real-time signal processing related problems. The knowledge of various transforms with the understanding of designing procedure of discrete systems will help students to work in multi-disciplinary fields of engineering.				
Syllabus Representation of Signals and Systems: Basic continuous and discrete time signals and systems, Energy and power signals, System modeling concepts, Discrete and continuous Linear Time Invariant (LTI) systems, Properties of LTI systems. Time and frequency characterization : Continuous and discrete time Fourier series and its properties, Continuous and discrete time Fourier transforms and its properties, Correlation, Autocorrelation, Transfer function, Solution of differential equation.. Z-Transform: Z-transform (ZT) and its properties, Region of Convergence, Inverse Z-transform - Power series, partial fraction expansion, residue method, Transfer function, Solution of difference equation, Relation between ZT and FT. Discrete Fourier Transforms (DFT): Introduction to DFT and its properties, Fast Fourier transforms (FFT), Decimation in time and decimation in frequency: Radix-2 FFT algorithm. Design of Discrete Time Systems: Introduction to Filters and Filtering, Difference between analog and digital filters. Design of Finite impulse response (FIR) Filters: Symmetric and Antisymmetric FIR filters, Linear phase concept, Design of FIR filter (LPF, HPF, BPF and BRFF) using window functions. Design of Infinite Impulse response (IIR) Filters: Butterworth approximation, Design of Butterworth Lowpass filters using approximation of Derivative, Impulse invariance and Bilinear Transformation, Difference between IIR and FIR filters. Implementation of Discrete Time Systems: Structure Realization for FIR filters: Direct-I, cascade, linear phase and lattice structures. Structure Realization for IIR filters: Direct-I, Direct-II, cascade, parallel and lattice structures.				
Course Learning Outcomes (CLOs) The students will be able to: 1. analyze different properties of continuous and discrete time signals and systems. 2. represent signals and systems in the frequency domain using Fourier tools. 3. apply Z-transform to analyze discrete time signals and system. 4. implement FFT algorithm. 5. design of Discrete Time Systems. 6. implementation of Discrete Time Systems.				
Text Books 1. Oppenheim, A.V. and Willsky, A.S., Signal & Systems, Prentice Hall of India (1997). 2. J.G. Proakis, D.G. Manolakis and D. Sharma, Digital Signal Processing, Pearson, 3rd edition, (2013).				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Reference Books

1. Kani, A.N. Signals and Systems, McGraw Hill Higher Education (2011)
2. Digital Signal processing-A Practical Approach, second edition, Emmanuel I. feacher, and Barrie W. Jervis, Pearson Education.
3. Digital Signal Processing, S. Salivahanan, A. Vallavaraj, C. Gnapriya TM.

Evaluation scheme

Sr.No	Evaluation Elements	Weights (%)
1.	MST	25
2.	EST	40
3.	Sessionals (May include Assignments/Projects/Tutorials/Quiz/Lab evaluations)	35

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